

IoT-Based Surveillance System Using Raspberry Pi

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Problem Statement:

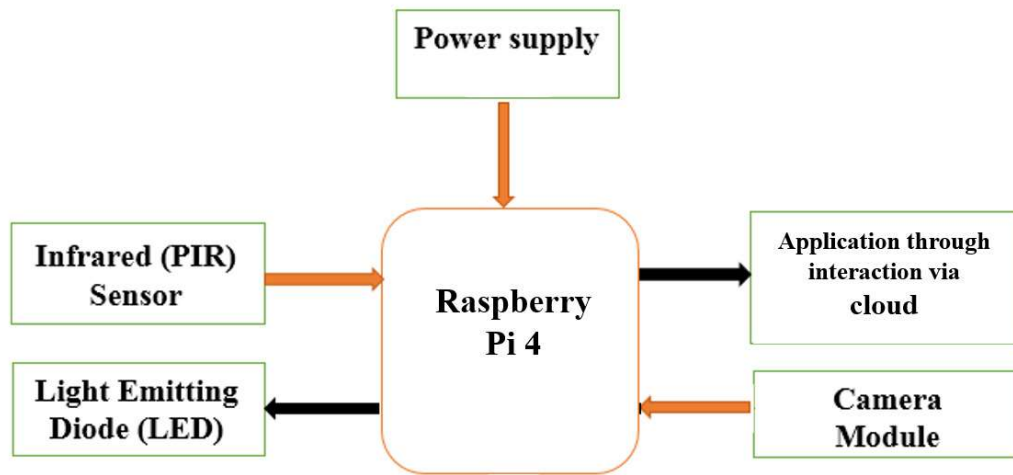
Traditional home surveillance systems face several challenges: high costs that limit accessibility, frequent false alarms caused by basic motion detection, and delays in alerting users about potential security threats. Additionally, the lack of real-time monitoring and limited access to captured data make these systems less effective in ensuring prompt action. Cloud-based solutions, while convenient, often raise concerns about user privacy due to indiscriminate data uploads. These limitations hinder the trust and usability of such systems. To overcome these challenges, a smarter, more efficient, and privacy-conscious surveillance system is required, capable of providing timely alerts, capturing relevant footage, and enabling secure access to stored data.

Aim:

This project aims to develop an intelligent, cost-effective IoT-based surveillance system using Raspberry Pi. The system will:

- Utilize a motion detection sensor to accurately trigger event recording and minimize false alarms.
- Automatically capture images during motion events and securely store them in the cloud.
- Send instant alerts to users via email with event details and images.
- Provide secure remote access to cloud-stored images while prioritizing user privacy by uploading only relevant data.

Basic Block Diagram:



Specification:

- HC-SR501 PIR Motion Sensor:
 - Voltage Range: DC 4.5V- 20V
 - Detection Distance: 3 to 7m
 - Work temperature: -20-+80°C
- Raspberry Pi Camera Module 2
 - Sensor: 8 MP Sony IMX219
 - Video Quality: 1080p30 FPS, 720p 60 FPS, 640x480p 60/90 FPS
 - Dimension: Around 25 × 24 × 9 mm

Cloud Used: SupaBase Open Source

Project Features:

1. Motion Detection

- Utilizes a PIR motion sensor (`gpiozero.MotionSensor`) to detect movement.
- Activates when motion is detected, making it ideal for security and monitoring systems.

2. Image Capture

- Integrates with a Raspberry Pi camera module to capture images (picam-still) when motion is detected.
- Timestamped filenames ensure unique identification of captured images.

3. Email Notifications

- Automatically sends an email to a predefined recipient when motion is detected.
- Email includes:
 - Subject: Indicates motion detection.
 - Body: Explains the purpose of the email.
 - Attachment: Captured image of the event.

4. Cloud Storage Integration (Supabase)

- File Upload: Captured images are uploaded to a Supabase storage bucket.
- Public URL: Generates a public URL for the uploaded image, enabling access and sharing.

5. Customizable Email System

- Supports Gmail SMTP for sending emails, but can be extended to work with other providers.

6. Real-time Monitoring

- Continuously monitors for motion and processes events in real-time.
- Outputs status updates to the console for monitoring activity (e.g., "Motion detected!" or "No motion detected.").

7. Automation and Timing

- Incorporates delays to prevent redundant captures and notifications within a short time frame.

8. Configurable Settings

- Variables for email credentials, recipient address, Supabase URL, and API key are customizable, enabling flexibility for different users.

Use Case:

- Home Security: Detect intrusions and notify homeowners.
- Wildlife Monitoring: Observe and record animal movements in specific areas.
- Workplace Safety: Monitor unauthorized access to restricted zones.

Code:

```
1  from gpiozero import MotionSensor
2  from time import sleep
3  import os
4  import smtplib
5  from email.mime.multipart import MIMEMultipart
6  from email.mime.text import MIMEText
7  from email.mime.base import MIMEBase
8  from email import encoders
9  from datetime import datetime
10 from supabase import create_client, Client
11
12 # Configure motion sensor
13 pir = MotionSensor(4)
14
15 # Email credentials
16 sender_email = "YOUR_EMAIL"
17 sender_password = "YOUR_PASSWORD"
18 recipient_email = "RECIPIENT_EMAIL"
19
20 # Supabase configuration
21 url = "URL"
22 key = "APIKEY"
23 supabase: Client = create_client(url, key)
24 bucket_name = "jimit-bucket"
25
26 print("Waiting for sensor to stabilize...")
27 sleep(30) # Wait for 30 seconds to let the PIR sensor stabilize
28 print("Sensor ready.")
29
30 def send_email(image_path):
31     try:
32         # Set up the MIME
33         msg = MIMEMultipart()
34         msg['From'] = sender_email
35         msg['To'] = recipient_email
36         msg['Subject'] = "Motion Detected - Image Attached"
37         body = "Motion was detected. Please find the captured image attached."
38         msg.attach(MIMEText(body, 'plain'))
39
```

```

40     # Attach the image file
41     with open(image_path, "rb") as attachment:
42         part = MIMEBase('application', 'octet-stream')
43         part.set_payload(attachment.read())
44         encoders.encode_base64(part)
45         part.add_header('Content-Disposition', f'attachment; filename={os.path.basename(image_path)}')
46         msg.attach(part)
47
48     # Connect to the Gmail server
49     with smtplib.SMTP('smtp.gmail.com', 587) as server:
50         server.starttls()
51         server.login(sender_email, sender_password)
52         server.sendmail(sender_email, recipient_email, msg.as_string())
53
54     print("Email sent successfully.")
55
56 except Exception as e:
57     print(f"Error: {e}")
58
59 def upload_to_supabase(image_path):
60     filename = os.path.basename(image_path)
61     with open(image_path, "rb") as file:
62         try:
63             # Upload the image to Supabase
64             result = supabase.storage.from_(bucket_name).upload(filename, file)
65
66             # Get the public URL of the uploaded image
67             public_url = supabase.storage.from_(bucket_name).get_public_url(filename)
68
69             print(f"Image uploaded successfully. Public URL: {public_url}")
70             return public_url
71         except Exception as e:
72             print(f"Error uploading image to Supabase: {e}")
73             return None
74
75 while True:
76     if pir.motion_detected:
77         print("Motion detected!")
78
79     # Generate timestamp and update the image path
80     timestamp = datetime.now().strftime("%d%m%y_%H_%M_%S")
81     image_path = f"/home/pi/Desktop/img_{timestamp}.jpg" # Ensure this path matches your device setup
82
83     # Capture an image using the camera
84     os.system(f"rpivid --raw -o {image_path}")
85     print("Image captured.")
86
87     # Send email with the captured image
88     send_email(image_path)
89
90     # Upload the image to Supabase
91     upload_to_supabase(image_path)
92
93     # Delay to avoid multiple captures and emails in a short time
94     sleep(15)
95 else:
96     print("No motion detected.")
97     sleep(5)

```

Working Prototype:



Conclusions:

The IoT-based surveillance system using Raspberry Pi provides a cost-effective and intelligent solution to traditional home security challenges. By integrating motion detection, instant email alerts, and secure cloud storage, it ensures timely response, minimizes false alarms, and offers remote access to captured images. Prioritizing user privacy through selective data uploads, the system is reliable, efficient, and scalable, addressing modern surveillance needs with simplicity and effectiveness.